

Improving on AAVDiff: A Technical Literature Review for Conditional Discrete Diffusion of AAV2 Capsid Sequences

TL;DR

- **Classifier-free guidance (CFG) is the right call and should beat AAVDiff's gradient-following classifier guidance** — but only if you adopt a *properly discrete* CFG formulation (Schiff et al. 2024, D-CFG) rather than naively adding continuous gradients; for short, small-vocabulary protein regions, the strongest recent results come from either uniform-noise diffusion (UDLM) with D-CFG or reward fine-tuning (DRAKES), and CFG buys fidelity at a measurable, fundamental cost in diversity that must be tuned.
- **Architecturally, keep AdaLN-Zero (it beats cross-attention and in-context conditioning in every DiT ablation), embed the scalar fitness score with a Fourier/sinusoidal map + MLP summed into the timestep embedding, switch the corruption process from D3PM's predefined Q_t to absorbing/masked or uniform-noise diffusion (MDLM/MD4/UDLM), and add structural grounding** (position encodings reflecting the 561–588 surface loop, charge/volume features, or ESM/ProteinMPNN context).
- **ESM-2 embeddings are worth trying but are not a guaranteed win** for a 28-residue hypervariable loop saturated with synthetic mutations — pretrained protein language models sometimes underperform one-hot on small, highly-mutated, single-region datasets; use frozen ESM features as an auxiliary input or as denoiser initialization (DPLM-style), not as the sole representation, and benchmark against learned embeddings.

Key Findings

1. Guidance. AAVDiff "follows the gradient direction to identify viable samples," a form of classifier/value guidance. The literature is now clear that (a) naive continuous classifier guidance is not mathematically valid on discrete states (you cannot add a continuous gradient to a discrete categorical objective), and (b) discrete-appropriate CFG and classifier guidance both work and generally improve controllability over autoregressive baselines. Schiff et al. (2024, "Simple Guidance Mechanisms for Discrete Diffusion Models," arXiv:2412.10193) derive discrete classifier-free guidance (D-CFG) and discrete classifier-based guidance (D-CBG), finding "classifier-free guidance is more useful when paired with diffusion models compared to AR." Nisonoff et al. (2024, "Unlocking Guidance for Discrete State-Space Diffusion," arXiv:2406.01572) provide exact continuous-time-Markov-chain classifier guidance. Both report a fidelity↔diversity Pareto tradeoff governed by guidance strength γ .

2. Conditioning mechanism. The DiT ablation (Peebles & Xie 2023, arXiv:2212.09748) is decisive: across four DiT-XL/2 variants — in-context (119.4 GFLOPs), cross-attention (137.6), adaLN (118.6), and adaLN-Zero (118.6) — "adaLN-Zero outperforms cross-attention and in-context conditioning at all stages of training," with "the FID achieved with the adaLN-Zero

model... nearly half that of the in-context model" at 400K iterations, while cross-attention adds ~15% FLOPs (137.6 vs ~118.6) and still loses. For a *continuous scalar* fitness score (not a class label), the supported recipe is: encode the scalar with a sinusoidal/Gaussian-Fourier feature map → small MLP (SiLU) → add to the timestep embedding → drive the shared AdaLN-Zero modulation MLP. CCDM (Ding et al. 2024, arXiv:2405.03546) is direct precedent for diffusion conditioned on continuous regression labels. (arXiv)

3. ESM embeddings. DPLM (Wang et al. 2024, arXiv:2402.18567) shows discrete diffusion *built on* ESM-2 weights is a superior representation learner and generator, and a two-stage (MLM-then-diffusion) recipe helps. But evidence is mixed for small, highly-mutated, single-region tasks: multiple fitness-prediction studies find ESM embeddings sometimes underperform one-hot/UniRep on small or short-protein datasets.

4. Biologically-grounded architecture. The 561–588 region is hypervariable region VIII, surface-exposed loop IV at the threefold-symmetry spike, overlapping the AAV2-unique heparan-sulfate motif (R585, R588) and antibody epitopes. This biology motivates: (a) structure-aware position encodings, (b) structure-conditioned sequence models (ProteinMPNN, ESM-IF, GRADE-IF), (c) explicit charge/volume features to capture the K/R/C-unfavorable observation, and (d) multi-subunit/epistasis context (the capsid is a 60-mer assembly). ProteinMPNN has been applied directly to AAV2 capsid redesign with functional, intact capsids.

5. Continuous-to-discrete. Three paradigms: genuinely discrete (D3PM, masked/absorbing MDLM/SEDD), continuous-relaxation with discretization (bit/analog-bits diffusion, Dirichlet/simplex DDSM), and Gumbel-softmax/straight-through. For short biological sequences, genuinely discrete diffusion is now dominant and avoids rounding/off-manifold failure modes.

6. Corruption process. Across discrete diffusion, absorbing/masking ("[MASK]") is consistently the best-performing corruption, with two key exceptions relevant here: small-vocabulary regimes (amino acids ≈ 20–21) where uniform noise is on par or better (UDLM), and protein-specific BLOSUM-structured transitions (EvoDiff, GRADE-IF) that bias corruption toward biochemically similar amino acids.

Details

Question 1 — Classifier-free guidance vs. AAVDiff's gradient-following approach

AAVDiff's described mechanism — "following the gradient direction to identify viable samples during the generation process" — is conceptually classifier/value guidance: a separate viability signal nudges generation. The literature gives several concrete reasons to expect CFG (done correctly) to be at least competitive and often better, plus important caveats.

Why naive classifier guidance is problematic on discrete states. As the SVDD authors (Li et al. 2024, arXiv:2408.08252) put it, a model like ESM3 "cannot be integrated with standard classifier

guidance, because adding a continuous gradient to a discrete objective is not inherently valid." AAVDiff's gradient-following therefore likely operates on logits/embeddings (like NOS), which is a heuristic. NOS (Gruver et al. 2023, "Protein Design with Guided Discrete Diffusion," NeurIPS, arXiv:2305.20009) formalizes this as following "gradients in the hidden states of the denoising network," enabling sequence-space design and generalizing Bayesian optimization (LaMBO-2). This is essentially a more principled version of what AAVDiff does and is a strong baseline to compare against. (arXiv + 2)

Discrete CFG and its empirical record. Schiff et al. (2024, arXiv:2412.10193) derive D-CFG by sampling each token from a tempered product of conditional and unconditional models, $p_{\theta}^{\gamma}(z_s|z_t, y) \propto p_{\theta}(z_s|z_t, y)^{\gamma} \cdot p_{\theta}(z_s|z_t)^{1-\gamma}$. Their headline conclusions: (1) "classifier-free guidance is more useful when paired with diffusion models compared to AR"; (2) their D-CBG is the best classifier-based method; (3) on small-vocabulary biological/chemical data their UDLM (uniform-noise) model is more amenable to guidance than masked diffusion, because once a masked token is unmasked it is "locked in," whereas uniform noise lets tokens be refined repeatedly. On species-conditioned genome generation, only UDLM improved as γ increased while AR and MDLM degraded. This last point is directly relevant: AAV HVR VIII has a ~21-token vocabulary (small), so uniform-noise diffusion + D-CFG is a strong, evidence-backed candidate. (Discrete-diffusion-guidance)

The fidelity-diversity tradeoff. The original CFG paper (Ho & Salimans 2022, arXiv:2207.12598) is explicit that guidance trades diversity for fidelity: increasing strength "place[s] probability mass farther away from other classes... most of the mass becomes concentrated in smaller regions," boosting quality metrics "at the expense of decreased diversity." Recent theory (arXiv:2509.22007) shows strong early guidance erodes global diversity while strong late guidance suppresses fine detail, motivating time-varying γ schedules. For AAVDiff's stated dual goal (viability AND diversity), this is the central tension: too-aggressive guidance will collapse the very diversity the project is trying to maximize. The "Steering Generative Models" paper (arXiv:2505.15093, 2025) maps fitness-diversity Pareto fronts for Continuous, D3PM, and MDLM generators after guidance on real protein landscapes (TrpB, CreiLOV), finding guided diffusion enriches high-fitness variants and beats an ARLM+DPO baseline, with larger γ pushing fitness up and diversity down. (arXiv + 2)

Reward fine-tuning as an alternative to inference-time guidance. DRAKES (Wang et al. 2024, ICLR 2025, arXiv:2410.13643) fine-tunes a discrete diffusion model end-to-end by backpropagating a reward through the sampling trajectory via the Gumbel-softmax trick, with KL regularization to a pretrained model to preserve "naturalness." On enhancer-activity and protein-stability design it outperforms classifier guidance (CG), SMC, TDS, and CFG baselines. The lesson: if you have a good differentiable viability oracle, RL/reward fine-tuning can beat inference-time guidance — at higher compute cost and risk of reward over-optimization (out-of-distribution sequences with low likelihood under the pretrained model). (arXiv)

Verdict on Q1: Yes, expect a properly-discrete CFG to outperform AAVDiff's gradient heuristic, with two strong refinements: (i) consider uniform-noise UDLN + D-CFG given the small amino-acid vocabulary; (ii) if a reliable viability predictor exists, benchmark reward fine-tuning (DRAKES-style) as the likely performance ceiling. Tune γ on an explicit fitness-diversity Pareto curve rather than picking one value.

Question 2 — Embedding and injecting the continuous fitness condition

AdaLN-Zero vs. alternatives. The DiT ablation (Peebles & Xie 2023, arXiv:2212.09748) trained four XL/2 variants — in-context (119.4 GFLOPs), cross-attention (137.6), adaLN (118.6), adaLN-Zero (118.6) — and found "adaLN-Zero outperforms cross-attention and in-context conditioning at all stages of training," with "the FID achieved with the adaLN-Zero model... nearly half that of the in-context model" at 400K iterations. Zero-initialization (each block starts as identity) is the critical ingredient; the follow-up "Unveiling the Secret of adaLN-Zero in Diffusion Transformers" (OpenReview E4roJSM9RM) confirms via ablation that "the impact of zero-initialization is particularly notable, leading to significant performance improvement," that "adding [the] scaling element α is beneficial as well," and proposes adaLN-Gaussian for further gains. Caveat: for *free-form* text conditioning, cross-attention can beat adaLN (arXiv:2312.04557), but a single scalar fitness value is the easy case adaLN-Zero handles best. So: keep AAVDiff's AdaLN mechanism — it is the correct choice — but make it AdaLN-Zero if it isn't already. (OpenReview)

How to embed a scalar. The supported pipeline mirrors how DiT injects class labels, swapping the class lookup table for a continuous encoder: scalar fitness $s \rightarrow$ sinusoidal/Gaussian-Fourier feature embedding $\rightarrow$ 2-layer MLP with SiLU $\rightarrow$ add to the timestep embedding $\rightarrow$ shared AdaLN-Zero MLP producing per-block scale/shift/gate. The Fourier step matters: Tancik et al. (2020, "Fourier Features Let Networks Learn High Frequency Functions," NeurIPS, arXiv:2006.10739) show that "passing input points through a simple Fourier feature mapping enables a multilayer perceptron (MLP) to learn high-frequency functions in low-dimensional problem domains," because "a standard MLP has impractically slow convergence to high frequency signal components" and the Fourier map overcomes this spectral bias via "a stationary kernel with a tunable bandwidth." Score-SDE/EDM diffusion codebases already use Gaussian Fourier embeddings for the noise level, so the same module transfers directly to the fitness scalar. CCDM (Ding et al. 2024, arXiv:2405.03546) is the most direct precedent for diffusion conditioned on continuous regression labels and adds "vicinal" training to cope with label sparsity plus a regression-adapted CFG — a useful trick given continuous log2 selection scores are unevenly sampled. (arxiv)

Bio-sequence precedent for continuous conditioning. DDSM (Avdeyev et al. 2023, ICML, arXiv:2305.10699) conditions promoter generation on a continuous CAGE transcription-initiation signal by concatenating the signal as an extra input channel; NOS conditions on a continuous fitness value via a value/regression head and hidden-state gradients. Both are alternatives to AdaLN injection worth knowing, but for a single global scalar over a short sequence, AdaLN-Zero is simpler and stronger.

Question 3 — ESM / protein language model embeddings

The strongest pro-ESM evidence is DPLM (Wang et al. 2024, arXiv:2402.18567): a discrete diffusion model initialized from ESM-2 that is both a better generator and "a superior representation learner... comparing favorably to ESM2," and whose two-stage MLM→diffusion training beats training the diffusion objective from scratch (DPLM-FS). DPLM-2 extends this to joint sequence-structure. This suggests initializing your denoiser from ESM-2 weights (rather than only using ESM as a frozen feature extractor) could help.

However, transfer is not guaranteed for this specific task. Reasons grounded in the literature: (a) ESM-2 is trained on *natural* proteins, while Bryant et al. (2021, Nat Biotechnol 39:691–696) generated viable variants carrying "12–29 mutations across this region" (with up to 19 substitutions or 15 insertions in the 28-residue segment, and 58.1% — 106,665 — forming viable capsids) — far outside natural sequence density, exactly the synthetic, highly-mutated regime where evolutionary priors may mislead. (b) Multiple fitness studies report ESM underperforming simpler encodings on small/short datasets: one affibody study explicitly "wondered why ESM underperformed One-Hot and UniRep," attributing it to small protein length and complex fitness. (c) Medium-sized ESM-2 models and specific intermediate layers (not the last layer) often transfer best, so layer choice matters. (d) The HVR VIII loop is short with limited evolutionary context — the regime where PLM priors are weakest. [PubMed](#) [Nature](#)

Verdict on Q3: Try ESM-2 as (i) initialization for the denoiser (DPLM-style two-stage training) and (ii) an auxiliary frozen-feature input, but *always* benchmark against learned-from-scratch embeddings and one-hot, and sweep which ESM layer to extract. Do not assume ESM is better; on this saturated single-region task it may not be.

Question 4 — Biologically-grounded architectural improvements

The biology: residues 561–588 are AAV2 VP1 hypervariable region VIII, the surface-exposed loop IV at the threefold-axis spike. R585 and R588 form the heparan-sulfate proteoglycan (HSPG) binding motif unique to AAV2 (R484, R487, K532, R585, R588 collectively mediate HSPG binding, with R585/R588 spanning *different* VP subunits); the region overlaps antibody epitopes. Basic/charged or bulky residues (K/R/C) disrupting this loop plausibly impair assembly or receptor binding — consistent with AAVDiff's observation that K/R/C are unfavorable in HVR VIII. Concrete, grounded improvements: [ASM Journals](#) [PubMed](#)

- **Structure-aware position encodings / structural context.** Inject 3D information about where each residue sits on the capsid surface. Structure-conditioned sequence models are mature: ProteinMPNN (Dauparas et al. 2022, message-passing inverse folding), ESM-IF1 (Hsu et al. 2022), and discrete-diffusion inverse folding GRADE-IF (Yi et al. 2023, NeurIPS, arXiv:2306.16819), which runs discrete diffusion over amino acids on an SE(3)-equivariant EGNN conditioned on backbone structure and notably uses BLOSUM as its transition kernel "to encode the prior knowledge of the response of AAs to evolutionary pressures."

MapDiff (arXiv:2412.07815) combines an EGNN structure predictor with a masked sequence designer. (arXiv)

- **AAV-specific structural ML precedent.** ProteinMPNN has been applied directly to AAV2 capsid redesign (Jiang, Laosinwattana & Dalby, UCL, bioRxiv 2025, 10.1101/2025.07.24.666527): they "targeted 52% of AAV2 residues for redesign by ProteinMPNN, such that only the buried protein core and subunit interfaces would be varied, leaving the capsid external and internal surface features unchanged," using PDB 1LP3 fivefold/threefold interfaces at sampling temperature 0.1; final designs were "between 14% and 30% mutated overall, and yet were capable of forming functional and intact capsids." This demonstrates that structure-conditioned design works for AAV and validates multi-subunit interface awareness. (Society + 3)
- **Multi-subunit / epistasis modeling.** The capsid is a 60-mer; HSPG sites (R585/R588) span *different* VP subunits, so viability depends on inter-subunit context a single-sequence model ignores. This argues for at least encoding the structural neighborhood, even approximately.
- **Pretrain-then-finetune.** EvoDiff and DPLM both show broad-protein pretraining then task fine-tuning helps; combine with the small AAV dataset.
- **Biophysical constraints.** Encode per-residue charge/volume as auxiliary features or use a BLOSUM-structured corruption (see Q6) so the model's noise model respects biochemistry — directly addressing the K/R/C-unfavorable observation.

Question 5 — Continuous-to-discrete / forcing discreteness

Three families:

(a) **Genuinely discrete diffusion.** D3PM (Austin et al. 2021, arXiv:2107.03006) with categorical transition matrices; the modern, simpler, better-performing instances are absorbing/masked diffusion — MDLM (Sahoo et al. 2024, arXiv:2406.07524) and MD4 (Shi et al. 2024) — and score-based SEDD (Lou et al. 2024, ICML best paper, arXiv:2310.16834). SEDD "beats existing language diffusion paradigms (reducing perplexity by 25-75%) and is competitive with autoregressive models, in particular outperforming GPT-2," generating with "around 6-8× better generative perplexity than un-annealed GPT-2" and "similar quality with 32× fewer network evaluations." These never leave the discrete simplex, so they have *no rounding error and no off-manifold problem*. This is AAVDiff's family and remains the safest choice for short protein sequences. (arxiv) (arxiv)

(b) **Continuous relaxation + discretize at the end.** Bit/analog-bits diffusion (Chen et al. 2022, arXiv:2208.04202) encodes tokens as real-valued "analog bits" then thresholds; self-conditioning is essential to get clean bimodal outputs. Simplex/Dirichlet approaches — DDSM (Avdeyev et al. 2023, ICML) and Dirichlet Flow Matching (Stark et al. 2024, arXiv:2402.05841) — diffuse in the probability simplex. Latent approaches (DiscDiff) diffuse in a learned latent then decode. The DNA-design community notes these "risk introducing distortions during reconstruction,"

whereas directly modeling token transitions (discrete) avoids this — a relevant warning for a short, information-dense region. (University of Toronto) (OpenReview)

(c) **Gumbel-softmax / straight-through / argmax rounding.** Used mainly to make discrete sampling differentiable for reward backprop (DRAKES uses the Gumbel-softmax trick precisely for this), not usually as the core generative mechanism.

Verdict on Q5: For a 28-residue region, stay in genuinely discrete diffusion (absorbing/masked or uniform). The continuous-relaxation routes add rounding/reconstruction failure modes for little benefit at this length. Reserve Gumbel-softmax for optional reward fine-tuning.

Question 6 — Corruption process / noise design

The empirical consensus across discrete diffusion is that **absorbing/masking corruption is best** in most settings: MDLM states "absorbing state (i.e., masking) diffusion consistently achieves the best performance," echoed by MD4, SEDD, and others. D3PM's original options were uniform, absorbing, and discretized-Gaussian (ordered by similarity); for images D3PM-Gauss beat absorbing/uniform, but for sequences absorbing/masking generally wins. (S-sahoo)

Two protein-specific refinements matter here:

- **BLOSUM-structured transitions.** EvoDiff (Alamdari et al. 2023) implements both D3PM-Uniform and a biologically-informed D3PM-BLOSUM that corrupts amino acids toward biochemically similar ones; GRADE-IF uses BLOSUM as its transition kernel. Interestingly, EvoDiff found its order-agnostic masking (OADM) variant *outperformed* both uniform and BLOSUM D3PM and was the only variant whose performance scaled with data — a caution that BLOSUM structuring is not automatically better.
- **Small-vocabulary uniform noise.** Schiff et al. (2024) showed that "in small vocabulary regimes, uniform noise is on par or better than absorbing state," and uniform noise is more guidable (tokens can be revised). With ~21 amino-acid tokens, this is squarely the small-vocab regime. (NSF PAGES)

Noise schedules. D3PM used cosine for uniform and a mutual-information / $(T-t+1)^{-1}$ schedule for absorbing. The cosine schedule (from MaskGIT) "slow[s] down the unmasking process at the beginning," reducing simultaneous conflicting unmaskings; it generally improves sample quality and convergence over linear.

Verdict on Q6: Two well-supported options. Option A (controllability-first): uniform-noise UDLM with cosine schedule + D-CFG — best for the small vocabulary and best for guidance. Option B (biochemistry-first): absorbing/masked diffusion (MDLM-style) optionally with a BLOSUM-informed transition or charge/volume-aware corruption to respect the K/R/C-unfavorable biology — but A/B test BLOSUM vs. plain masking, since EvoDiff found masking can win.

Stage 1 — De-risk the guidance change (highest expected value). Replace AAVDiff's gradient-following with discrete classifier-free guidance (Schiff et al. D-CFG). Train a conditional denoiser on the continuous Bryant log2 selection score plus an unconditional model (via condition dropout). Embed the scalar with a Fourier-feature + MLP module summed into the timestep embedding, injected via AdaLN-Zero. Benchmark: produce a fitness-diversity Pareto curve over $\gamma \in \{1, 2, 3, 4\}$; success = strictly dominating AAVDiff's reported viability rate at matched diversity. If guidance collapses diversity, adopt a time-varying γ schedule (strong late, weak early).

Stage 2 — Fix the corruption process. A/B test three forward processes on held-out reconstruction and downstream viability: (i) AAVDiff's current D3PM Q_t , (ii) absorbing/masked MDLM with cosine schedule, (iii) uniform-noise UDLM with cosine schedule. Given the ~21-token vocabulary and the priority on guidability, expect (iii) to win; threshold to switch = better fitness-at-fixed-diversity than (i).

Stage 3 — Add biochemical/structural grounding. Add per-residue charge and volume as auxiliary input features (directly targets the K/R/C-unfavorable finding) and either (a) BLOSUM-informed transitions or (b) structure-aware position encodings keyed to the 561-588 surface-loop geometry. If a backbone is available, prototype a ProteinMPNN/ESM-IF or GRADE-IF structure-conditioned variant as an upper-bound comparator — ProteinMPNN already produces functional AAV2 capsids at 14-30% overall mutation.

Stage 4 — Try ESM, but gated on evidence. Initialize the denoiser from ESM-2 (DPLM two-stage MLM→diffusion) and/or add frozen ESM features; sweep extraction layer. Keep this only if it beats learned-from-scratch and one-hot on your validation viability — on a 28-residue saturated-mutation region it may not.

Stage 5 — If you have a strong viability oracle, test reward fine-tuning. Benchmark a DRAKES-style Gumbel-softmax reward backprop with KL regularization against your best CFG model. This is the likely performance ceiling but watch for reward over-optimization (monitor sequence log-likelihood under the pretrained model).

Benchmarks that should change the plan: if uniform-noise diffusion underperforms masking on your data, revert to MDLM (the small-vocab advantage is empirical, not guaranteed); if ESM features don't beat one-hot in cross-validation, drop them; if $\gamma > 1$ always collapses diversity below target, prefer reward fine-tuning with explicit diversity regularization over inference-time guidance.

Caveats

- **AAVDiff's exact guidance mechanism is under-documented.** "Following the gradient direction" is interpreted here as classifier/value guidance, but the paper's description is

vague; confirm against their code before claiming CFG is strictly different.

- **Most cited evidence is from text, images, DNA, or general proteins, not AAV capsids specifically.** Transfer to a 28-residue surface loop is an inference, not a measured result. The small-vocabulary uniform-noise advantage and the masking-vs-BLOSUM question should be settled empirically on your own data.
- **Recent AAV generative-diffusion preprints (e.g., a 2026-dated "AAVDiffusion" viability-guided latent diffusion bioRxiv) are non-peer-reviewed and report in-silico (classifier-predicted) viability only;** treat their claimed superiority cautiously and do not assume wet-lab validation.
- **The guidance fidelity-diversity tradeoff is fundamental, not a tuning artifact** — any viability gain from strong guidance will cost diversity, and AAVDiff's headline value is diversity, so this tradeoff is the crux of the project.
- **CFG requires training an unconditional model (or condition dropout); reward fine-tuning requires a trustworthy differentiable oracle.** Both add engineering/compute cost over AAVDiff's single-model approach.
- A couple of subagent-surfaced AAV/PLM references carried anomalous future-dated arXiv identifiers and were excluded from load-bearing claims.